

Estimated media enrichment conditions

Bin	Likely physiology	Estimated enrichment / media conditions	Explanation
97	Metabolically flexible. Slow growing. Syntrophic	- Simple organics at low level (e.g., small organic acids / simple sugars) - Ammonium as N source + small amount of amino acids or peptides as backup. - Sulfide or sulfur species as redox partner option. - Vitamin/cofactor mix. - Co-culture friendly conditions.	Glycolysis core + nucleotide and amino acid metabolism + cofactors/vitamins + plus energy metabolism signals (oxidative phosphorylation + methane metabolism genes + sulfur metabolism genes)
80	Strong heterotroph	Anaerobic heterotroph enrichment: - Simple carbon (sugars/organic acids) - Ammonium + optional peptides (casamino acids/yeast extract) - Keep reduced conditions with sulfide or cysteine as reductant	Complete glycolysis + PRPP + amino-acid related pathways. The presence of sulfur and methane metabolism pathways indicates adaptation to anaerobic conditions
58	Reduced metabolic breadth; more likely dependent on community metabolites	Richer, supportive anaerobic enrichment: - Peptides / amino acids (yeast extract/casamino acids). - Simple carbon - Vitamin mix. - Maintain anaerobic/reducing conditions	Fewer complete modules overall but still has glycolysis core + PRPP and limited biosynthesis. This pattern often shows dependency on external nutrients (vitamins & amino acids) and growth improves with complex organics.
51 (high contamination)	Uncertain	Supportive anaerobic enrichment as bin 58	No complete module in ghostkoala + high contamination. The genes may be mixed from multiple organisms, so any media design would be unreliable.

Tree

Tree scale: 0.1

